



External Quality Assessment for Detection of Fetal Trisomy 21, 18, and 13 by Massively Parallel Sequencing in Clinical Laboratories

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Accepted for publication
October 21, 2015.

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An external quality assessment for detection of trisomy 21, 18, and 13 by massively parallel sequencing was implemented by the National Center for Clinical Laboratories of People's Republic of China in 2014. Simulated samples were prepared by mixing fragmented abnormal DNA with plasma from non-pregnant women. The external quality assessment panel, comprising 5 samples from pregnant healthy women, 2 samples with sex chromosome aneuploidies, and 13 samples with different concentrations of fetal fractions positive for trisomy 21, 18, and 13, was then distributed to participating laboratories. In total, 55.6% (47 of 84) of respondents correctly identified each of the samples in the panel. Seventeen false-negative and 87 gray zone results were reported, most [102 of 104 (98.1%)] of which were derived from for trisomy samples with effective fetal fractions <4%. No laboratories generated false-positive results. In addition, we observed varied diagnostic capabilities of different assays, with the assay on the basis of NextSeq CN500 performing better than others, whereas Z values generated by BGISEQ-100 fluctuated greatly. There were no significant correlations between the numbers of unique sequence reads and Z values from any trisomy sample generated by BGISEQ-100. Overall, most clinical laboratories detected samples containing effective fetal fractions >4%. Our study shows need for further laboratory training in the management of samples with low fetal fractions. For some assays, precision of Z values needs to be improved. (*J Mol Diagn* 2016, ■: 1–9; <http://dx.doi.org/10.1016/j.jmoldx.2015.10.003>) ^{Q5}

^{Q6} Chromosomal aneuploidy is a common cause of birth defects, and trisomy 21 (T21), trisomy 18 (T18), and trisomy 13 (T13) are the most frequently occurring chromosomal abnormalities. Currently, the standard screening procedures for detecting chromosomal abnormalities involve a combination of first-trimester serum screening for biochemical markers, such as pregnancy-associated plasma protein-A, free β -human chorionic gonadotropin, and ultrasound measurement of the nuchal translucency thickness, with or without a measurement of second-trimester markers. Depending on the results of these screens, the definitive prenatal diagnosis, which includes amniocentesis or chorionic villus sampling, followed by chromosomal analysis, is performed in high-risk pregnancies. However, depending on the screening strategy used, the standard prenatal aneuploidy screening can only identify 50% to 95% of the

anomalies, with false-positive screen rates of 5%.^{1,2} Hence, the sensitivity of current screening programs for fetal chromosomal aneuploidy is unsatisfactory, and the high rates of false-positive results lead to unnecessary invasive sampling, which add stress to the pregnancies. In recent years, noninvasive prenatal testing (NIPT) using cell-free DNA (cfDNA) sequences isolated from maternal blood samples has been introduced into clinical practice for the detection of T21, T18, and T13. Several approaches have been reported for the analysis of chromosomal cfDNA fragments using next-generation DNA sequencing

Supported by the National Population and Family Planning Commission of the People's Republic of China Special Fund for Health-Scientific Research in the Public Interest grant 201402018. ^{Q2}

Disclosures: None declared. ^{Q3}

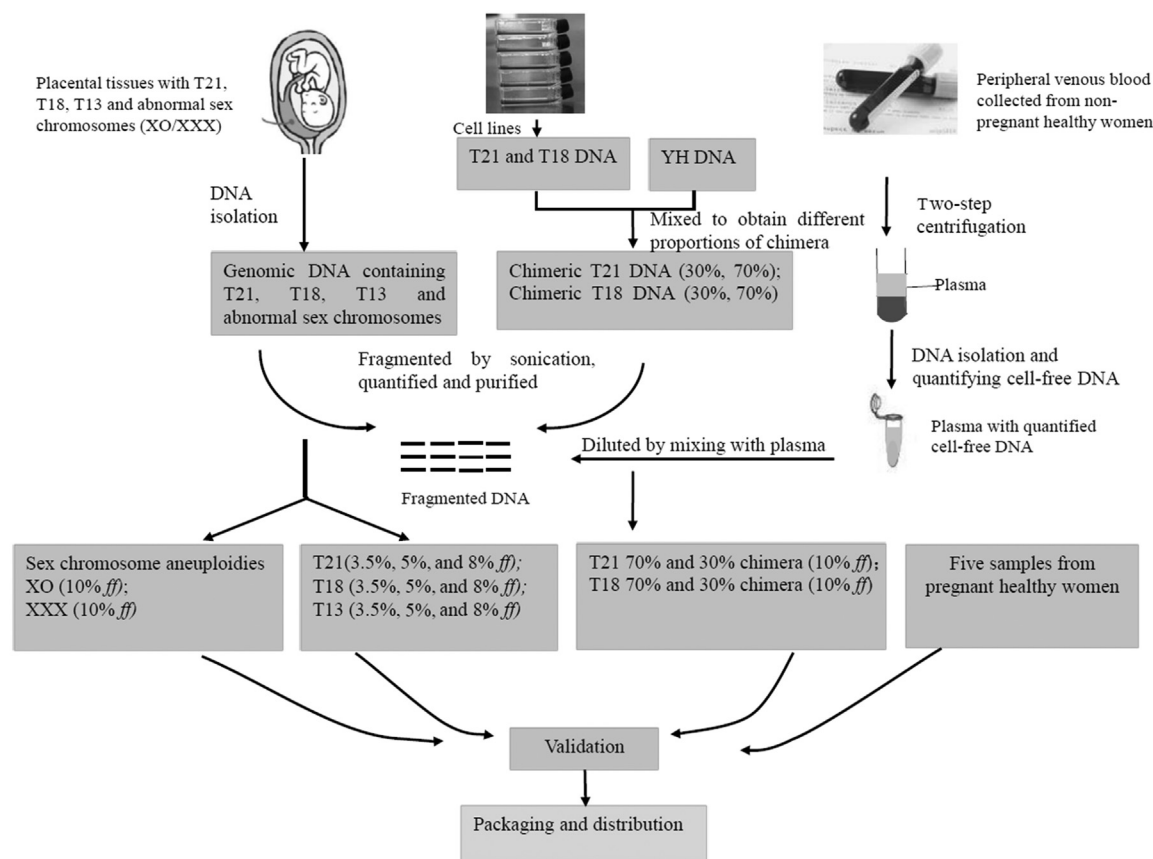


Figure 1 A flow diagram of the preparation, validation, and distribution of the samples in noninvasive prenatal testing external quality assessment scheme.

technology, including whole genome massively parallel sequencing (MPS),^{3–5} targeted MPS,^{6,7} and analysis of single-nucleotide polymorphisms.^{8,9} The results of multiple clinical trials of NIPT were promising and demonstrated high sensitivity (up to 99%) and specificity, with low false-positive rates (<0.1%).^{10–13} The International Society for Prenatal Diagnosis,¹⁴ the American College of Obstetricians and Gynecologists,¹⁵ and the American College of Medical Genetics and Genomics¹⁶ supported the proposal that pregnant women at high risk should be provided NIPT as a screening option for routine clinical care after receiving appropriate counseling.

As the testing transitioned from clinical trials to clinical care, NIPT has become commercially available. There are many reports regarding the use of NIPT in the clinical laboratories.^{17–19} Recently, in People's Republic of China, commercial vendors have simplified these bioinformatics analysis tools, and increasing clinical laboratories have shown interest in this new method for T21, T18, and T13 screening. Under the circumstances, external quality assessment (EQA) becomes an essential management measure to assess the proficiency and performance of various NIPT assays. It also ensures that laboratories maintain robust laboratory diagnostic capacities. However, it is hard to obtain enough clinical samples from pregnant

women having a fetus with trisomy. Therefore, we produced artificial specimens to simulate clinical samples. Herein, we described the first nationwide EQA study in People's Republic of China, organized by the National Center for Clinical Laboratories (Beijing, People's Republic of China), and discussed laboratory and assay performance for NIPT.

Materials and Methods

NIPT Panel Composition and Preparation

Panels were prepared by the National Center for Clinical Laboratories using placental tissues with chromosomal aneuploidy, cultured cell lines, YH DNA (First Asian Diploid Genome, <http://yh.genomics.org.cn>, last accessed September 21, 2015),²⁰ and peripheral venous blood from non-pregnant healthy women (Figure 1). First, genomic DNA (gDNA) was extracted from placental tissues with T21, T18, T13, and abnormal sex chromosomes (XO/XXX). Two cell lines (AG09802 and GM02732) with trisomies of chromosomes 21 and 13 were purchased from the Coriell Cell Repositories (Coriell, NJ). T21 and T18 DNAs were extracted from cultured cell lines using the DNA mini kit (Qiagen, Venlo, the Netherlands) and mixed with YH

Table 1 Panel Composition and Performance of 84 Laboratories for NIPT Testing

Sample no.	Classification	Trisomy 21		Trisomy 18		Trisomy 13	
		No. of correct data sets/total no. of data sets (%)	No. of gray zone data sets/total no. of data sets (%)	No. of correct data sets/total no. of data sets (%)	No. of gray zone data sets/total no. of data sets (%)	No. of positive data sets/total no. of data sets (%)	No. of gray zone data sets/total no. of data sets (%)
1401	Normal chromosome	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1402	8% T21	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1403	3.50% T18	84/84 (100)	0	66/84 (78.6)	18/84 (21.4)	84/84 (100)	0
1404	XXX positive	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1405	8% T13	84/84 (100)	0	84/84 (100)	0	83/84 (98.8)	1/84 (1.2)
1406	10% T18–70% chimera	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1407	Normal chromosome	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1408	8% T18	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1409	Normal chromosome	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1410	10% T21–30% chimera	62/84 (73.8)	20/84 (23.8)	84/84 (100)	0	84/84 (100)	0
1411	10% T18–30% chimera	84/84 (100)	0	44/84 (52.4)	26/84 (31.0)	84/84 (100)	0
1412	Normal chromosome	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1413	5% T13	84/84 (100)	0	84/84 (100)	0	83/84 (98.8)	1/84 (1.2)
1414	5% T21	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1415	10% T21–70% chimera	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1416	Normal chromosome	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1417	3.50% T13	84/84 (100)	0	84/84 (100)	0	62/84 (73.8)	21/84 (25.0)
1418	5% T18	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1419	3.50% T21	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0
1420	XO positive	84/84 (100)	0	84/84 (100)	0	84/84 (100)	0

NIPT, noninvasive prenatal testing.

DNA to obtain different proportions of chimeric T21 and chimeric T18 DNA. gDNA was quantified using a FLUOstar Omega plate reader (BMG LABTECH, Ortenberg, Germany), fragmented with a focused ultrasonicator (Covaris, Inc., Woburn, MA), and then purified in a two-step process using AMPure XP beads (Agencourt Bioscience Corp., Beverly, MA). The expected size distribution of the resulting DNA fragments was 100 to 200 bp, which was confirmed by agarose gel electrophoresis, and fragmented DNA was quantified [value A (ng/μL)] using a FLUOstar Omega plate reader (BMG LABTECH). Second, samples of peripheral venous blood were collected from non-pregnant healthy women using EDTA tubes. Plasma was then separated from the blood samples by a two-step centrifugation method: whole blood was centrifuged at $1600 \times g$ for 10 minutes at 4°C , and the supernatants were transferred to 2.0-mL tubes. Samples were then centrifuged again under $16,000 \times g$ for 10 minutes in 4°C , and supernatants were transferred into clean bottles for storage and then allocated to 50-mL BD tubes (Becton Dickinson, Franklin Lakes, NJ). All plasma samples were pooled to obtain mixed plasma, and all procedures were completed within 4 hours. Cell-free DNA was extracted from 600 μL of mixed plasma using the Extraction and Purification Kit for Human Peripheral Blood Genomic DNA (BGI-Wuhan, Wuhan, People's Republic of China), and DNA was quantified

with a FLUOstar Omega plate reader (BMG LABTECH). The total concentration of 30 replicate samples was recorded, and the average was used as the content of cfDNA in 600 μL mixed plasma [value B (ng/μL)]. Last, the fragmented gDNA samples extracted from placental tissues were diluted by mixing with plasma to be used for preparation of 20 samples. Table 1 summarizes the composition of the panel. The panel comprised 5 samples from pregnant healthy women, 2 samples with sex chromosome aneuploidies, and 13 samples positive for T21, T18, and T13. To simulate samples positive for chromosomal abnormalities, gDNA extracted from the different abnormal placental tissues or cell lines was chosen to reflect various abnormal cell-free fetal DNA, such as T21, T18, T13, T21 chimera, and T18 chimera. In addition, different amounts of gDNA were used to reflect various fetal fraction (*ff*) levels that could be present in clinical samples, including 3.5%, 5%, and 8% *ff*. The volumes of fragmented gDNA to be added into 50-mL BD tubes were calculated according to the following formula: $A \times C / (A \times C + B \times E) = D$, where *D* was the desired *ff* to be simulated, *C* was the total volume of fragmented gDNA, and *E* was the volume of plasma. After production of each of the 20 samples, the preparations were dispensed in 700-μL volumes into 1.5-mL siliconized vials. The vials were labeled by the National Center for Clinical Laboratories NIPT 2014 and were

Table 2 Information of Each Assay Involved in the EQA

		No. of data sets	Minimum library concentration	Minimum no. of raw sequence data	Minimum no. of unique sequence reads	Thresholds of Z values		
Commercial providers	Sequencing platforms					Positive	Gray zone	Negative
BGI	BGISEQ-100 (OEM of Ion Proton)	21	≥0.15 ng/μL	>5M	>3.2M	>4	1.96~4	<1.96
BGI	BGISEQ-1000 (OEM of CG)	3	≥3 ng/μL	>6M	>2M	>4	1.96~4	<1.96
Daan	DA8600 (OEM of Ion Proton)	3	>0.200 ng/μL	NR	≥3.0M	>3.11	2.39~3.11	<2.39
Berry Genomics	NextSeq CN500 (OEM of NextSeq 500)	39	≥10 pM	>3.5M	>1.8M	≥3	NA	<3
CapitalBio	BioelectronSeq 4000 (OEM of Ion Proton)	10	≥1 nmol/L	NR	≥3.0M	>3	2~3	<2
Illumina	NextSeq 500	1	≥2 ng/μL	≥10M	≥4M	0 > 3.5	2.5~3.5	-2.5~2.5
		1	>3 pM	>5M	≥3M	>3	1.96~3	<1.96
		1	>2 ng/μL	≥10M	≥4M	>3.5	2.8~3.5	-2.8~2.8
		1	>1 nmol/L	NR	>3M	>3	2~3	<2
5 ABI	Ion Proton	1	>0.20 ng/μL	NR	≥3M	>3	NA	<2
1		≥0.2 ng/μL	NR	>2M	>3	NA	<3	
1		≥20 nmol/L	≥6M	≥4M	>4	2~4	<2	
Illumina	Hiseq2000	1	>5 ng/μL	≥6M	>5M	>3.5	2.5—3.5	<2.5
Illumina	Hiseq2500	1						

EQA, external quality assessment; NA, not applicable; NR, no requirement.

randomly assigned numbers from 1 through 20. The panels were stored at -70°C before shipment to the participating laboratories.

Validation of the NIPT Panel

The NIPT panel was evaluated by the Shenzhen Birth Defect Screening Project Laboratory (BGI-Shenzhen, Shenzhen, People's Republic of China) and by Berry Genomics (Beijing, People's Republic of China) with the same processes as their routine patient sample testing.

The BGI-Shenzhen group used a BGISEQ-100 sequencer (BGI-Wuhan). DNA samples were extracted from the NIPT panel using a Nucleic Acid Purification Kit (BGI-Wuhan) and then used for the construction of libraries using a Detection Kit for Noninvasive Fetal Trisomy (T21, T18, and T13; BGI-Wuhan), according to the manufacturer's instructions. The libraries were quantified using a Bioanalyzer 2100 (Agilent Technologies, Santa Clara, CA), and 10 libraries with different labels were pooled and quantified. The pooled library was sequenced using a Universal Reaction Kit for Sequencing (BGI-Wuhan) and a BGISEQ-100 sequencer.

The Berry Genomics group used a NextSeq CN500 sequencer (Berry Genomics-Hangzhou, Hangzhou, People's Republic of China). DNA was extracted using the Cell-Free DNA Extraction Kit (Berry Genomics-Hangzhou) and constructed into libraries with a Libraries Construction and Purification Kit (Berry Genomics-Hangzhou). The Step-OnePlus (Applied Biosystems, Carlsbad, CA) and Qubit 2.0 Fluorometer systems (Life Technologies, Carlsbad, CA) were used for the quantification and validation of the genomic libraries. Ninety-six libraries with different tags were pooled and quantified, including 20 samples in the

NIPT panel and clinical samples from their routine work. The pooled library was sequenced with a NextSeq CN500 sequencer using standard single-end 36 reads.

Participants and Data Analysis

Coded test panels of frozen samples were shipped on dry ice to 85 clinical laboratories. All laboratories were assigned the same coded samples and were required to use their routine procedures. Detailed instructions for storage conditions and assay procedures were provided. Because the samples contained fragmented gDNA by sonication, end repair was needed in assay procedure, which was different from routine test. Laboratories were asked to submit their results of testing for fetal trisomy, Z values, and the threshold of Z values for positivity online within 2 weeks of the receipt of the test panel. To calculate Z values, the number of reads aligned to the reference genome in NIPT was counted and compared with that of the same chromosome of a reference pool of euploid pregnancy samples. The Z value of a chromosome represents the number of SDs away from the reference mean percentage of representation of that chromosome.^{3,21} A Z value more than threshold indicates a statistically significant deviation between the sample and reference for the chromosome. Questionnaires were also sent to obtain information regarding the procedures used (the platforms and reagents used for DNA extraction, protocols for construction, and sequencing of DNA libraries) and the assay-specific quality control (QC) criteria (minimum total DNA per sample, minimum library concentration, and minimum number of unique sequence reads alignable to the genome). All of the results that differed from the reference were considered unacceptable. All data analyses were performed using SPSS version 16.0. Q8

Table 3 Performance of Different Assays for Each Trisomy Sample

Assay	No. of correct data sets/total no. of data sets of trisomy 21 (%)					No. of correct data sets/total no. of data sets of trisomy 18 (%)					No. of correct data sets/total no. of data sets of trisomy 13 (%)		
	Sample 1402	Sample 1410	Sample 1414	Sample 1415	Sample 1419	Sample 1403	Sample 1406	Sample 1408	Sample 1411	Sample 1418	Sample 1405	Sample 1413	Sample 1417
BGISEQ-100	21/21 (100)	5/21 (23.8)	21/21 (100)	21/21 (100)	21/21 (100)	16/21 (76.2)	21/21 (100)	21/21 (100)	1/21 (4.78)	21/21 (100)	21/21 (100)	21/21 (100)	5/21 (23.8)
BGISEQ-1000	3/3 (100)	0	3/3 (100)	3/3 (100)	3/3 (100)	2/3 (66.7)	21/21 (100)	21/21 (100)	0	21/21 (100)	3/3 (100)	3/3 (100)	3/3 (100)
DA8600	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)
NextSeq CN500	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)	39/39 (100)
BioelectronSeq 4000	10/10 (100)	10/10 (100)	10/10 (100)	10/10 (100)	10/10 (100)	3/10 (30)	10/10 (100)	10/10 (100)	0	10/10 (100)	10/10 (100)	10/10 (100)	10/10 (100)
NextSeq 500	3/3 (100)	2/3 (66.7)	3/3 (100)	3/3 (100)	3/3 (100)	0	3/3 (100)	3/3 (100)	0	3/3 (100)	3/3 (100)	3/3 (100)	0
Ion Proton	3/3 (100)	66.7 (2/3)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	3/3 (100)	1/3 (33.3)	3/3 (100)	3/3 (100)	3/3 (100)	2/3 (66.7)
Hiseq2000	1/1 (100)	0	1/1 (100)	1/1 (100)	1/1 (100)	0	1/1 (100)	1/1 (100)	0	1/1 (100)	1/1 (100)	1/1 (100)	0
Hiseq2500	1/1 (100)	1/1 (100)	1/1 (100)	1/1 (100)	1/1 (100)	0	1/1 (100)	1/1 (100)	0	1/1 (100)	0	0	0

Comparison of quantitative data was made using the *t*-test. Correlations of variables were evaluated with Spearman correlation analysis. $P < 0.05$ was regarded to be statistically significant.

Results

Validation of NIPT Panel

Samples containing effective *ff* > 3.5% (including 70% chimera samples containing 10% *ff*) were reported as positive by both BGI-Shenzhen and Berry Genomics. The T21 and T18 samples containing 3.5% *ff* were also correctly identified by the two laboratories. Meanwhile, the T13 sample containing 3.5% *ff* was initially recorded as gray zone by BGI-Shenzhen but was confirmed to be positive after retesting of the sample. Furthermore, 30% T21 and T18 chimera samples were counted as negative by BGI-Shenzhen. In contrast, Berry Genomics correctly identified all of the trisomy samples using NextSeq CN500 sequencer. No false-positive results were reported by either of the laboratories.

Panel Distribution and Response

Each of the 85 laboratories, including 73 clinical/hospital laboratories and 12 commercial laboratories, reported results before the closing date. A data set from one of the clinical laboratories did not include all of the results for the panel. As a result, the data from this laboratory were not included in the analysis presented herein. Each laboratory used

massively parallel sequencing to test the panel. The most common assay used was Berry Genomics NextSeq CN500 [39 of 84 (46.4%)], followed by BGISEQ-100 [21 of 84 (25.0%)], BioelectronSeq 4000 (CapitalBio, Beijing, People's Republic of China) [10 of 84 (11.9%)], BGISEQ-1000 (BGI-Wuhan) [3 of 84 (3.6%)], DA8600 (Daan, Guangzhou, People's Republic of China) [3 of 84 (3.6%)], NextSeq 500 (Illumina, San Diego, CA) [3 of 84 (3.6%)], Ion Proton (Applied Biosystems, Carlsbad, CA) [3 of 84 (3.6%)], HiSeq 2000 (Illumina) [1 of 84 (1.2%)], and HiSeq 2500 (Illumina) [1 of 84 (1.2%)]. QC requirements and the thresholds of each assay are summarized in Table 2. All of the laboratories declared that the results were reported when QC passed. The thresholds of most assays, except those of NextSeq CN500 and of one assay using the Ion Proton platform, contain gray zones.

NIPT Performance Testing

The T21 samples containing *ff* values of 8%, 5%, and 3.5% and the T18 samples containing *ff* values of 8% and 5% were correctly identified by all 84 laboratories (Table 1). The T18 sample containing 3.5% *ff* was correctly identified by 78.6% of the data sets, whereas 21.4% of laboratories reported gray zone. All of the laboratories, except one laboratory using Hiseq 2500, which reported gray zone, correctly identified the T13 samples containing 8% and 5% *ff*. Meanwhile, the T13 sample containing 3.5% *ff* was correctly identified by 73.8% of the data sets, with 25.0% of laboratories reporting gray zone and 1.2% of laboratories reporting negative. The T21 and T18 samples containing 10% (70% chimera) *ff* were also correctly

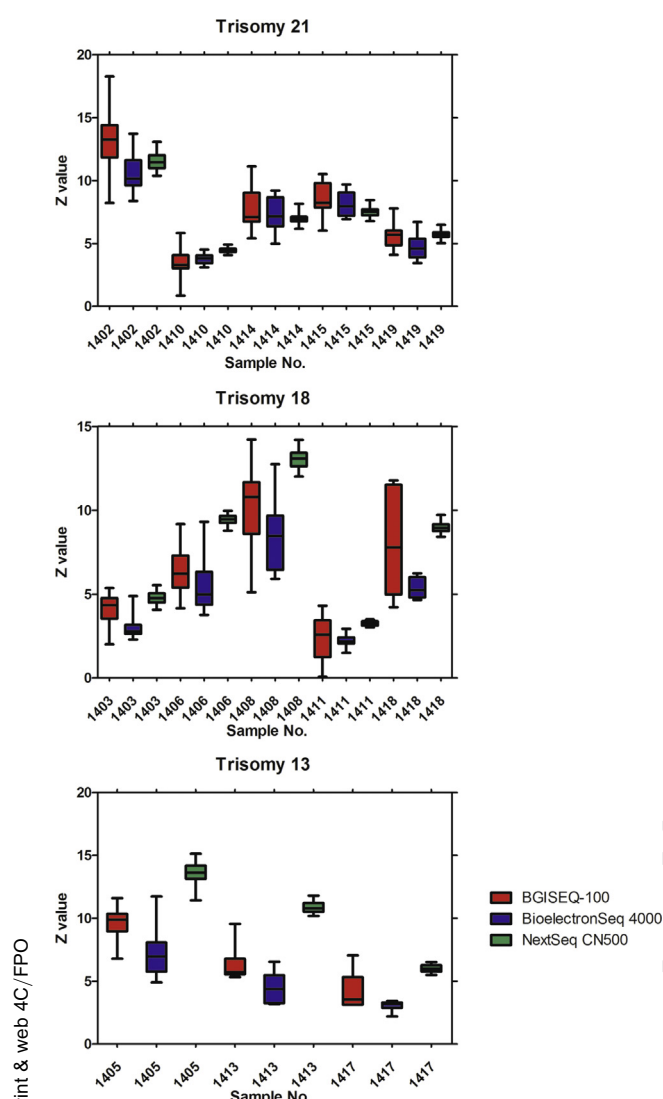


Figure 2 The Z values generated for trisomy samples in noninvasive prenatal testing panel by assays on the basis of BGISEQ-100, Bioelectron Seq 4000, and NextSeq CN500. The top, middle, and bottom lines in each box represent the 75th, 50th, and 25th percentiles of Z values, respectively. The error bars represent the maximum and minimum Z values.

identified in all of the data sets. Conversely, the T21 sample containing 10% (30% chimera) *ff* was correctly identified in 73.8% of the data sets, whereas 23.8% of laboratories reported gray zone and 2.4% of laboratories reported negative. The T18 sample with 10% (30% chimera) *ff* was correctly identified in 52.4% of the data sets, with 31.0% of laboratories reporting gray zone and 16.7% of laboratories reporting negative. Five samples from pregnant healthy women and the two samples with sex chromosome aneuploidies were correctly identified by all of the laboratories.

In total, only 55.6% (47 of 84) of the respondents correctly identified all of the samples in the panel. No false-positive results were reported by any of the 84 laboratories; however, 17 false-negative and 87 gray zone results were reported, most [102 of 104 (98.1%)] of which were derived from trisomy samples with effective *ff* <4%. For the T21,

T18, and T13 samples, 0.48% (2 of 420), 3.3% (14 of 420), and 0.4% (1 of 252) of the results were false negatives, respectively. In addition, 4.8% (20 of 420), 10.5% (44 of 420), and 9.1% (23 of 252) of the results for these samples were reported as gray zone, respectively.

Table 3 summarizes the performance levels of the different assay systems used. NextSeq CN500, which was the most widely used assay in this QC study, correctly identified all of the positive samples. Each of the other assays used, except the assay on the basis of HiSeq 2500, generated correct data sets for the T21, T18, and T13 samples containing effective *ff* >4%.

We next compared Z values generated by the assays used by >10 laboratories (Figure 2). For tests detecting T18 and T13, NextSeq 500 generated higher Z values than BGISEQ-100 and BioelectronSeq 4000. The Z values of the three assays were similar when T21 was detected. Z values generated by BGISEQ-100 fluctuated greatly. The numbers of unique sequence reads reported by the laboratories using the NextSeq CN500 were much higher than BGISEQ-100 and BioelectronSeq 4000 ($P < 0.0001$) (Figure 3). There were no significant correlations between the numbers of unique sequence reads and Z values from any sample generated by NextSeq 500, BGISEQ-100, and BioelectronSeq 4000 ($P > 0.05$).

Discussion

Multiple clinical validation studies presented high sensitivities, specificities, and negative predictive values for detection of T21, T18, and T13 by NIPT. These findings resulted in NIPT, by sequencing of cfDNA, being introduced as a method for clinical prenatal care. However, the EQA system for NIPT by next-generation DNA sequencing was not available. Ideally, the samples should be obtained from plasmas of pregnant women, but it is not feasible because of the limited amount of clinical samples. Herein, we prepared simulated samples by mixing fragmented abnormal gDNA with the plasma from non-pregnant women. Studies have previously shown that 80% of fetal-derived DNA has a size range of <193 bp,²² and a portion of fetal DNA molecules in plasma are <150 bp.²³ The size distribution of the fragmented DNA in our artificial samples was 100 to 200 bp, which includes most of the sizes of fetal fraction.²⁴ Both whole-genome and targeted MPS determine fetal ploidy state by comparing the ratio of cfDNA fragments from particular chromosomes in a test sample/reference values from euploid pregnancy samples.^{21,24,25} The Z value was then calculated to represent statistical deviations in the number of counts and ploidy state for that chromosome. Therefore, the synthetic samples are suitable for NIPT on the basis of MPS, such as MaterniT21 Plus,²⁶ Verifi,²⁷ Harmony Prenatal Test,²⁵ and all assays in our study.

There were also three limitations. First, the NIPT samples do not contain the fragments <100 bp, which could not assess the true efficiency of the testing laboratories in

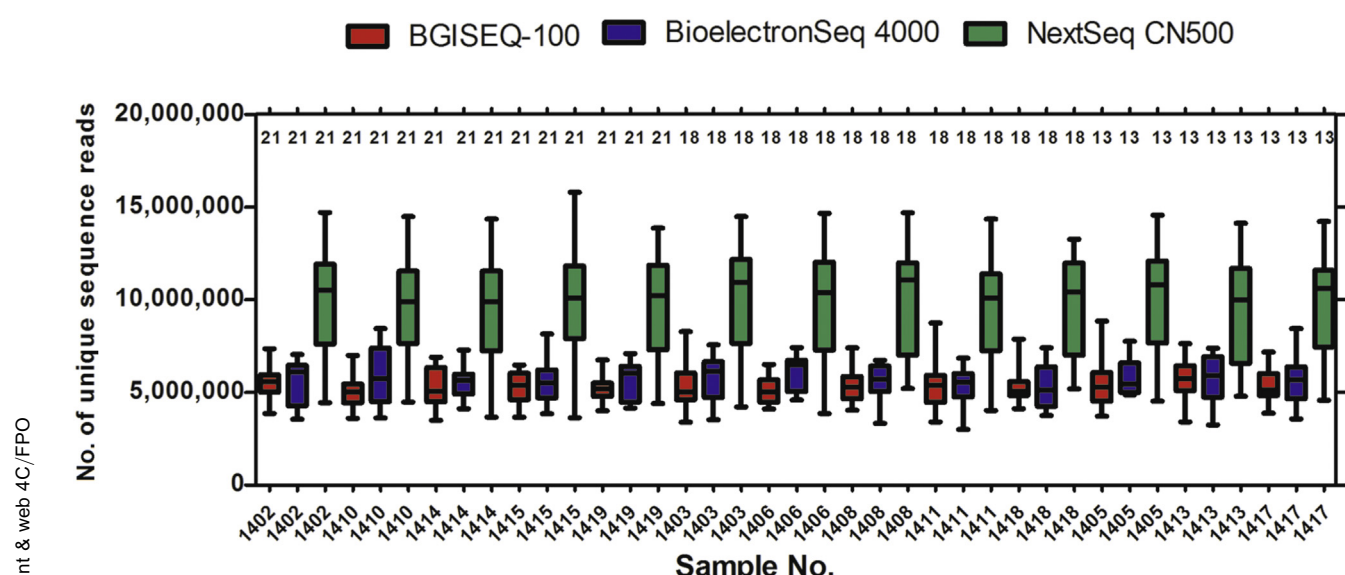


Figure 3 The numbers of unique sequence reads generated for the trisomy samples in noninvasive prenatal testing panel by assays on the basis of BGISEQ-100, Bioelectron Seq 4000, and NextSeq CN500. The top, middle, and bottom lines in each box represent the 75th, 50th, and 25th percentiles of unique sequence reads, respectively. The error bars represent the maximum and minimum unique sequence reads. Numbers below the plot represent the chromosomal aneuploidies detected.

extracting short fetal DNA molecules. In addition, the distribution is different between random fragmented DNA and the cell-free DNA from apoptotic cells.²⁴ Second, the DNA fragments in the plasma of non-pregnant women used in the study are shorter than those in the plasma of pregnant women.²² Third, the samples are not applicable for laboratories using the single-nucleotide polymorphism method, which use maternal genotypic information and recombination frequencies to construct possible fetal genotypes.²⁸ Then, each hypothesis is compared with actual plasma measurements, and a relative likelihood for each hypothesis is calculated. Because the artificial fetal DNA is not genetically related to the background plasma DNA in our artificial samples, NIPT on the basis of the single-nucleotide polymorphism method, like Panorama Prenatal Test,²⁸ is not suitable for our panel. However, given that all of the laboratories performed NIPT by MPS in this EQA, artificially fabricated samples are suitable to assess the performance of different workflows in individual laboratories. For T21 and T18 chimera samples in the EQA panel, cell lines were used, which were readily available and well-characterized.²⁹ Notably, it is important to confirm the status of cells by karyotypic analyses when new gDNA samples are prepared, because numerical aberrations in chromosomes might occur occasionally during the passage of cell lines.³⁰

Overall, most laboratories participating in this study have reliable diagnostic capacities for NIPT tests. No false-positive results were reported by any of the 84 laboratories. Problems mainly existed in false-negative results, all of which were caused by the failure to consistently identify the samples containing effective $ff < 4\%$. According to previous clinical trials and routine clinical testing, the minimum ff needed for reliable NIPT results is regarded as

approximately 4%,^{10,31} which is consistent with the results obtained by our analysis. Many factors can affect the ratio of fetal fraction, including delay of plasma separation, DNA extraction, gestation age, and overweight pregnant women.^{31,32} There are several approaches to quantify the ff for all pregnancies (ie, both male and female fetus), such as methods on the basis of locus-specific DNA methylation,³³ MPS,³⁴ and real-time quantitative PCR.³⁵ However, quantification of ff is not routinely performed in clinical laboratories of People's Republic of China, and clinical laboratories have no idea about whether the fetal DNA is adequate. Minimum ff should be another important QC parameter,^{36,37} and to measure ff would be helpful for laboratories to interpret the results when ff is lower than the limit of detection. We also noticed that T18 and T13 samples were correctly identified by fewer participants compared with T21 samples, which might correlate with GC-content bias. Chromosomes 13 and 18 both have a relatively lower GC percentage compared with chromosome 21; thereby, the CVs of read counts were higher.³ Thus, the sensitivities for T18 and T13 tend to be lower than T21.³⁸ Several GC-bias-correction models have been proposed to improve their sensitivities.^{39,40} In addition, DNA copy number losses or deletions in the maternal plasma could theoretically cause false-negative results,⁴¹ although no relevant clinical cases have been reported. Therefore, the biological reason of possible DNA copy number losses or deletions on chromosome 18 or 13 in plasma of non-pregnant women used could not be ruled out.

Noticeably, 4.8% (20 of 420), 10.5% (44 of 420), and 9.1% (23 of 252) of the results for the T21, T18, and T13 samples, respectively, were reported as gray zone. In some assays,²⁷ gray zone was used to address the samples

unclassified and those results were followed by karyotype analysis. Other assays, such as the MaterniT21 PLUS assay, gave a straightforward threshold.²⁶ In our study, the gray zone results were suggested to retest the sample according to the manuals for some assays. For example, the assays on the basis of the BGISEQ-100 platform recommend that if the Z value for a particular sample is between 1.96 and 4, the sample should be retested. If the Z value of the second test is ≥ 3 , the sample should be reported as positive. Conversely, if $Z < 3$, the sample should be reported as negative. However, for the samples with low ff , retests could not lead to much improvement. Figure 2 showed that Z values of samples 1410, 1411, and 1417, reported by BGISEQ-100, varied in a large range. Therefore, retests could also miss the positive samples in a portion of laboratories according to the distribution of Z values. It might be better for the women whose results are gray zone to receive further genetic counseling, comprehensive ultrasound evaluation, and diagnostic testing.⁴²

The laboratories using the NextSeq CN500 performed best and correctly identified all of the positive samples in the panel. NextSeq 500 generated much higher Z values than the threshold of 3. We also noticed that all of the CVs of Z values generated by BGISEQ-100 and BioelectronSeq 4000 were higher than NextSeq CN500 for the trisomy samples. Benn and Cuckle³⁶ showed that SDs of measured ff for chromosome 21 were smaller for high unique reads, which would improve the precision of Z values and the performance of the tests. Therefore, higher aligned unique sequence reads generated by NextSeq 500 might contribute to better performance of the platform. However, Z values were not associated with the numbers of aligned reads for positive samples detected by any of the three assays in our study. So, factors influencing the correct read representation, like incorrect assignment of fragments because of sequencing errors or polymorphisms, GC-content bias introduced by PCR,⁴³ and mappability of short reads to the repeat-rich regions of the genome,⁴⁴ should be also essential for fetal aneuploidy detection.

In conclusion, most clinical laboratories could detect samples containing effective $ff > 4\%$. Our study showed requirement for further laboratory training in the management of samples with low ff . For some assays, the precision of Z values needs to be improved. In our EQA for NIPT, detailed data analysis for the results was provided, so that all of the participants were aware about the performance of various methods and laboratories. We also provided the opportunity to retest samples if laboratories wanted to analyze their results. In the future, EQA for NIPT will be performed twice a year. The participants will be informed of the limitations of the test panel beforehand.

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